

# EMS Dashboard

Connected

Supervisory Python EMS with six scenarios, manual scenario control, manual price input, virtual battery charge, hydrogen estimation and live INA measurements in V, mA and mW.

## Demo and EMS control

EMS: started

Start the EMS system first. Then start the demo to reset the 96-slot timeline, begin CSV logging and activate the variable load trigger. Raw logging can run independently of the demo.

Bridge: OK

Reset

Start EMS system

Stop EMS system

Start demo + logging

Stop demo

Auto mode

Manual mode

Start raw log

|                    |                      |
|--------------------|----------------------|
| Raw log<br>Stopped | Raw log file<br>-    |
| Raw samples<br>0   | Last raw update<br>- |

### Price test date

25 / 04 / 2026

Load prices for date

### Manual electricity price

0.25707

Apply manual price

Hydrogen estimation (%)

100.0

Initiate H2 estimation manually

Virtual battery charge state (%)

90.0

Apply virtual battery state

Control mode  
**manual**

Demo status  
**Stopped**

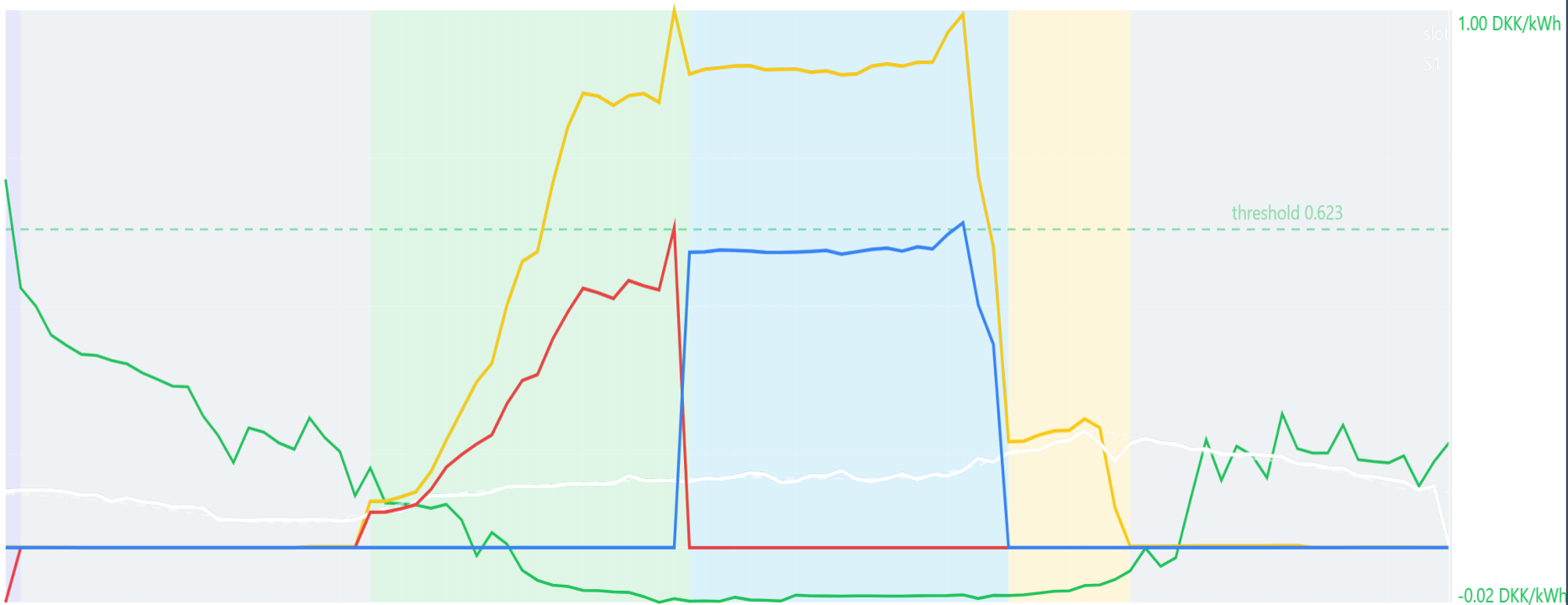
Current slot  
**95**

Scenario  
**S1**

System state timeline

One plot showing the selected price curve, demand profile and measured INA power in mW for PV, load, battery and PEM. Scenario background is shown where a slot has been logged.

- Price
- Price threshold
- Demand
- PV power
- Load power
- Battery power
- PEM power



Time  
23:45

Interval  
23:45-00:00

Price  
0.25707 DKK/kWh

Demand  
35.0 mW

PV power  
0.1 mW

Load power  
-0.0 mW

Battery power  
**-0.1 mW**

PEM power  
**-0.1 mW**

Virtual battery charge  
**90.0 %**

Hydrogen estimation  
**100.0 %**

Actual scenario  
**S1**

Current EMS decision

One 96-slot demo cycle was completed. Demo logging was stopped, the variable load trigger was disabled, and the system was returned to safe S1.

|                 |                 |
|-----------------|-----------------|
| EMS system      | Started         |
| Target scenario | S1              |
| Actual scenario | S1              |
| Reject reason   |                 |
| Price state     | LOW             |
| Current price   | 0.25707 DKK/kWh |

Manual scenarios

Selecting a scenario switches the EMS to manual mode and sends the scenario once.

S1 Grid → Load

S2 Grid → Load, PV → Battery

S3 Grid → Load, PV → PEM

S4 PV → Load

S5 Battery → Load

S6 PEM → Load

|                 |                                                   |
|-----------------|---------------------------------------------------|
| Current demand  | 35.0 mW                                           |
| PV available    | No                                                |
| Log file        | /app/python/logs/ems_demo_20260512_122830.csv     |
| Last log update | Demo cycle completed at 2026-05-12T12:52:31+02:00 |
| Error           | -                                                 |

Storage states

|                                   |                         |
|-----------------------------------|-------------------------|
| Current battery voltage           | 3.696 V                 |
| Initial SoC lookup voltage        | 3.693 V                 |
| Lookup-estimated real battery SoC | 4.7 %                   |
| Integrated real battery SoC       | 4.8 %                   |
| Integrated real battery charge    | 95.712 mAh / 2000.0 mAh |
| Virtual battery SoC               | 90.0 %                  |
| Virtual battery capacity          | 10.00 mAh               |
| Virtual battery charge            | 9.005 mAh               |
| Virtual battery state             | control_full            |

INA sensor measurements

|                 |          |
|-----------------|----------|
| PV voltage      | 2.725 V  |
| Battery voltage | 3.696 V  |
| PEM voltage     | 1.256 V  |
| Load voltage    | 5.123 V  |
| PV current      | 0.03 mA  |
| Battery current | -0.03 mA |
| PEM current     | -0.04 mA |
| Load current    | -0.01 mA |
| PV power        | 0.07 mW  |

|                     |          |
|---------------------|----------|
| Hydrogen volume     | 15.60 mL |
| Usable hydrogen     | 8.00 mL  |
| Hydrogen estimation | 100.0 %  |
| Hydrogen mode       | idle     |

|               |                 |
|---------------|-----------------|
| Battery power | -0.09 mW        |
| PEM power     | -0.05 mW        |
| Load power    | -0.04 mW        |
| Load trigger  | LOW             |
| Arduino mode  | S1 Grid -> Load |